

Normalization Assignment

Assignment: HR Database Normalization

Scenario

A company stores employee HR data in one spreadsheet.

emp_id	emp_name	department	manager	skills	certifications	project	project_location
E01	Aung Aung	IT	U Hla	Python, SQL	AWS, Azure	HRIS	Yangon
E02	Su Su	Finance	Daw May	Excel, Power BI	CPA	Payroll	Mandalay
E03	Kyaw Kyaw	IT	U Hla	Java, Python	AWS	HRIS	Yangon

Part A: Questions

Q1. What normalization problem exists in the original HR table?

Q2. Convert the HR table into 1NF.

Q3. What is the main issue after converting to 1NF?

Q4. Convert the 1NF table into 2NF.

Q5. What issue still exists after 2NF?

Q6. Convert the 2NF design into 3NF.

Q7. Is the HR database now in BCNF?

Q8. What 4NF issue can exist in HR data?

Q9. What is a possible 5NF issue in HR data?

Q10. What is the final normalized HR database design?